Project Report

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2.1 Project: Rides & Bookings Ingestion and Geospatial Processing

Step1 → Data Acquisition & Preparation

Step 1: Data Acquisition

Source: Kaggle – Uber NYC Dataset
Location: **https://www.kaggle.com/datasets/yzhoukaggle/zone-coordinates-of-nyc-yellow-taxi-trip-records**

The dataset includes trip IDs, start and end latitude/longitude, timestamps, and distance metrics, providing a realistic foundation for simulating ride data.

Step 2: Data Preparation

Original File Location: Local machine or downloaded folder (Data.xlsx)
Converted File Location: HDFS or project workspace (Data.parquet)
The **Excel (.xlsx) file** was converted to **Parquet** format because PySpark handles Parquet more efficiently, enabling better performance, schema management, and scalability for large-scale data processing. This prepared dataset served as the input for the data pipeline and geospatial analysis.

```
from pyspark.sql import SparkSession
import pandas as pd

spark = SparkSession.builder.appName("ExcelToParquet").getOrCreate()

pdf = pd.read_excel("/user/student/staging_zone/Data.xlsx")

df = spark.createDataFrame(pdf)

df.write.mode("overwrite").parquet("/user/student/staging_zone/Data.parquet")
```

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Step2 → Load To HDFS

```
[student@localhost Documents]$ hdfs dfs -put /home/student/Documents/Data_parque.parquet /staging_zone/
[student@localhost Documents]$ hdfs dfs -ls /staging_zone/
Found 3 items
-rw-r--r--   1 student supergroup      91437 2025-10-04 10:28 /staging_zone/Data.xlsx
-rw-r--r--   1 student supergroup      40025 2025-10-04 14:04 /staging_zone/Data_parque.parquet
drwxr-xr-x   - student supergroup          0 2025-10-04 11:47 /staging_zone/Staging_Rides_Geo
[student@localhost Documents]$
```

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Step3 → Transformation with PySpark

Step 1: Initialize Spark Session

- Start a Spark session with Hive support.

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("Transformations").enableHiveSupport().getOrCreate()
```

Step 2: Read Raw Data

- Read Parquet File and Store it in DataFrame

```
df = spark.read.parquet("/user/student/staging_zone/Data.parquet")
df.show(5)
df.printSchema()
```

Step 3: Initial Data Analysis

- Check latitude/longitude and datetime columns.
- Identify invalid records: trip_distance <= 0, start == end, coordinates = 0.

```
df.select("start_longitude","start_latitude","end_longitude","end_latitude").show(1001)
df.filter("trip_distance <= 0.0").show()
df.select("start_datetime","end_datetime").show(1001)
```

Step 4: Data Cleaning

1. Remove trips with zero or negative distance.

```
df = df.filter("trip_distance > 0.0")
```

1. Remove records with coordinates equal to 0.

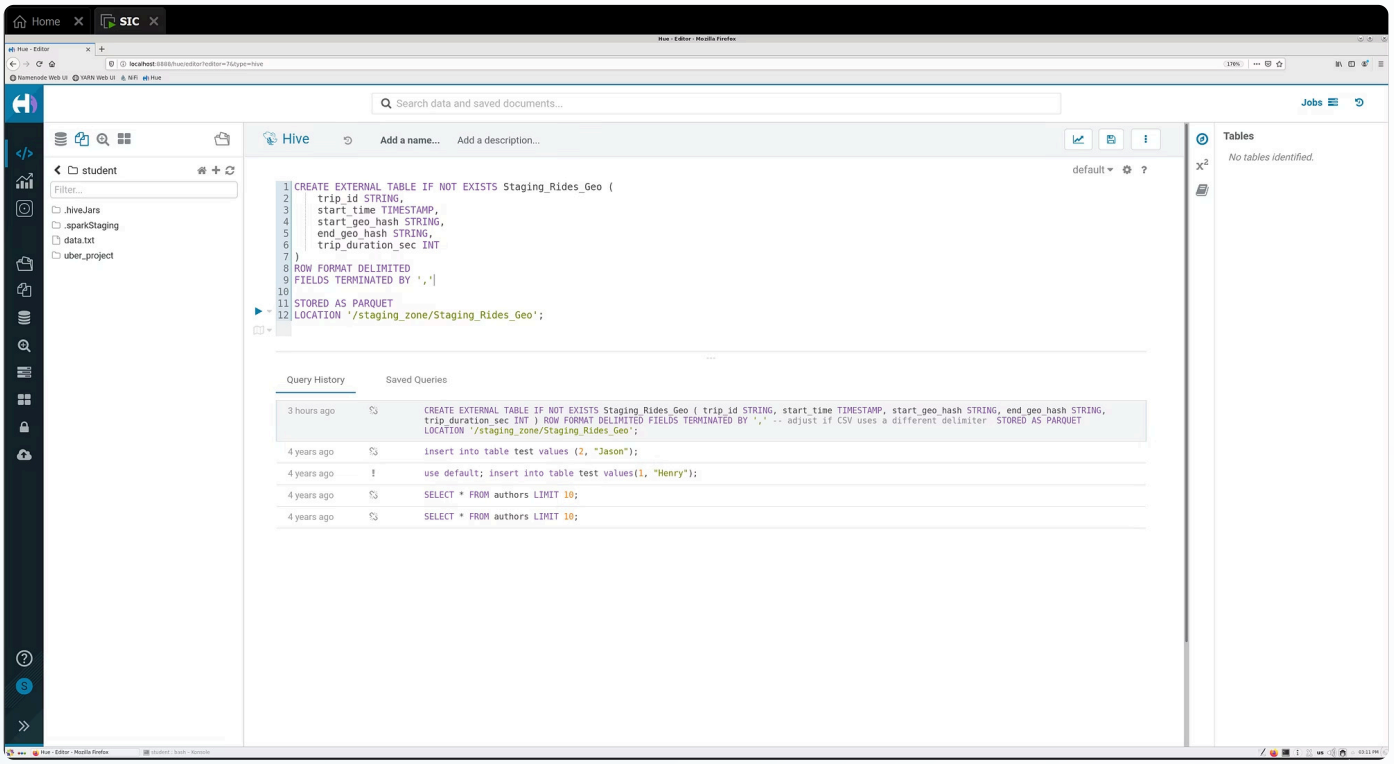
```
df = df.filter("start_longitude != 0.0 AND start_latitude != 0.0 AND end_longitude != 0.0 AND
end_latitude != 0.0")
```

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Step 4: Load to Hive, Explore, and Partition

Step 1: Create External Table in Hive

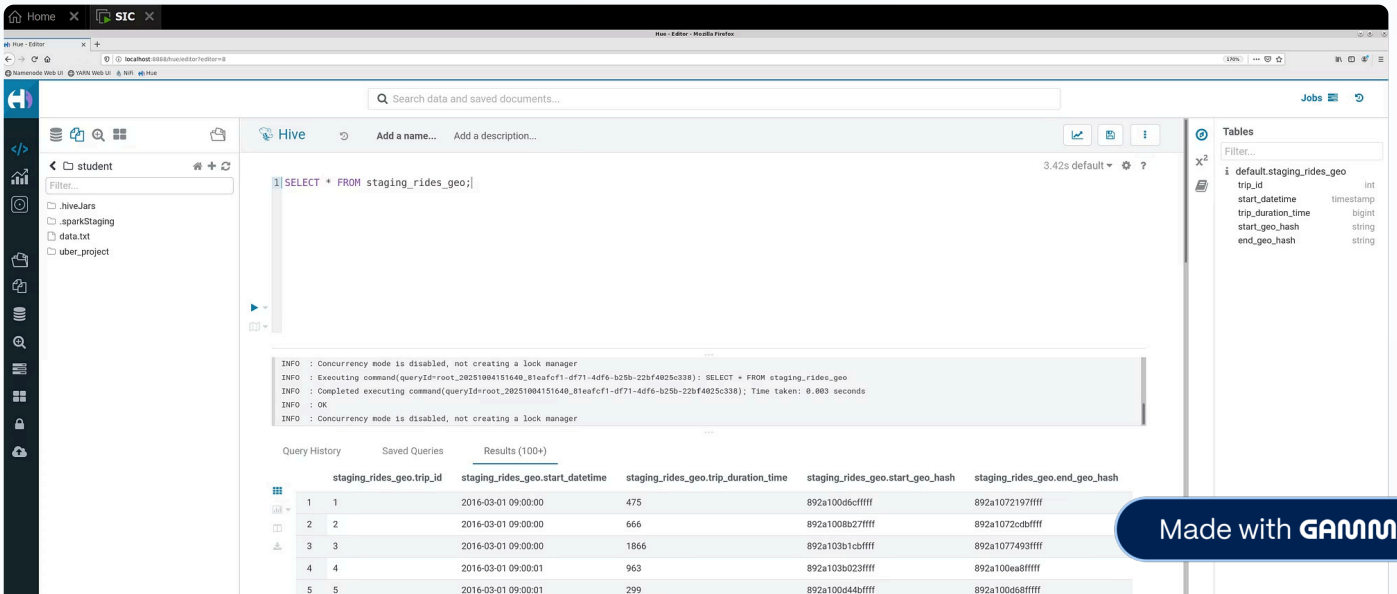
1. After transforming and cleaning the data in PySpark, the final DataFrame (`Tripes`) was written to Hive.
2. An **external table** was created so that the Hive table references the data stored in HDFS (or workspace), allowing reuse without moving the underlying files.



Step 2: Explore Data & Generate Insights

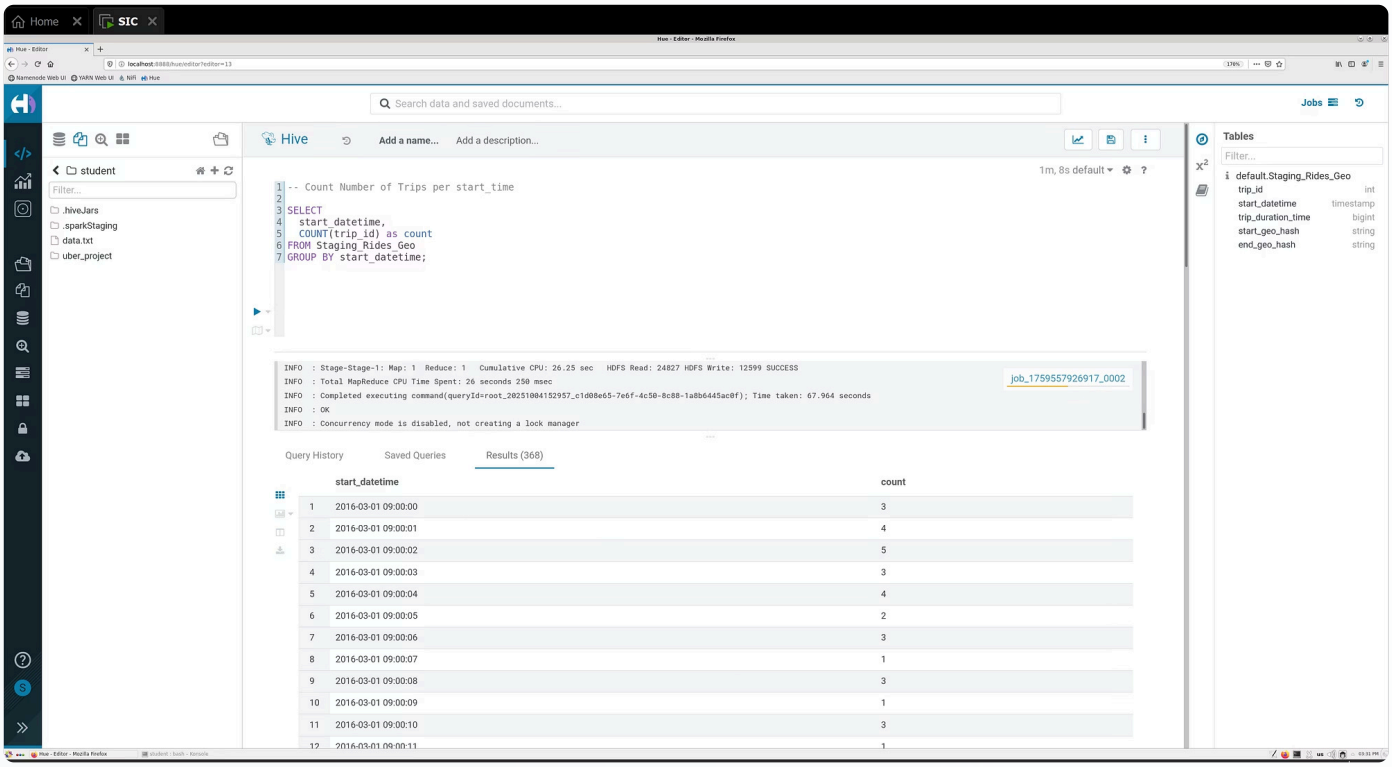
After creating the table, various **analytical queries** were executed in Hive to understand the dataset:

1. **Select all data** to inspect the table:

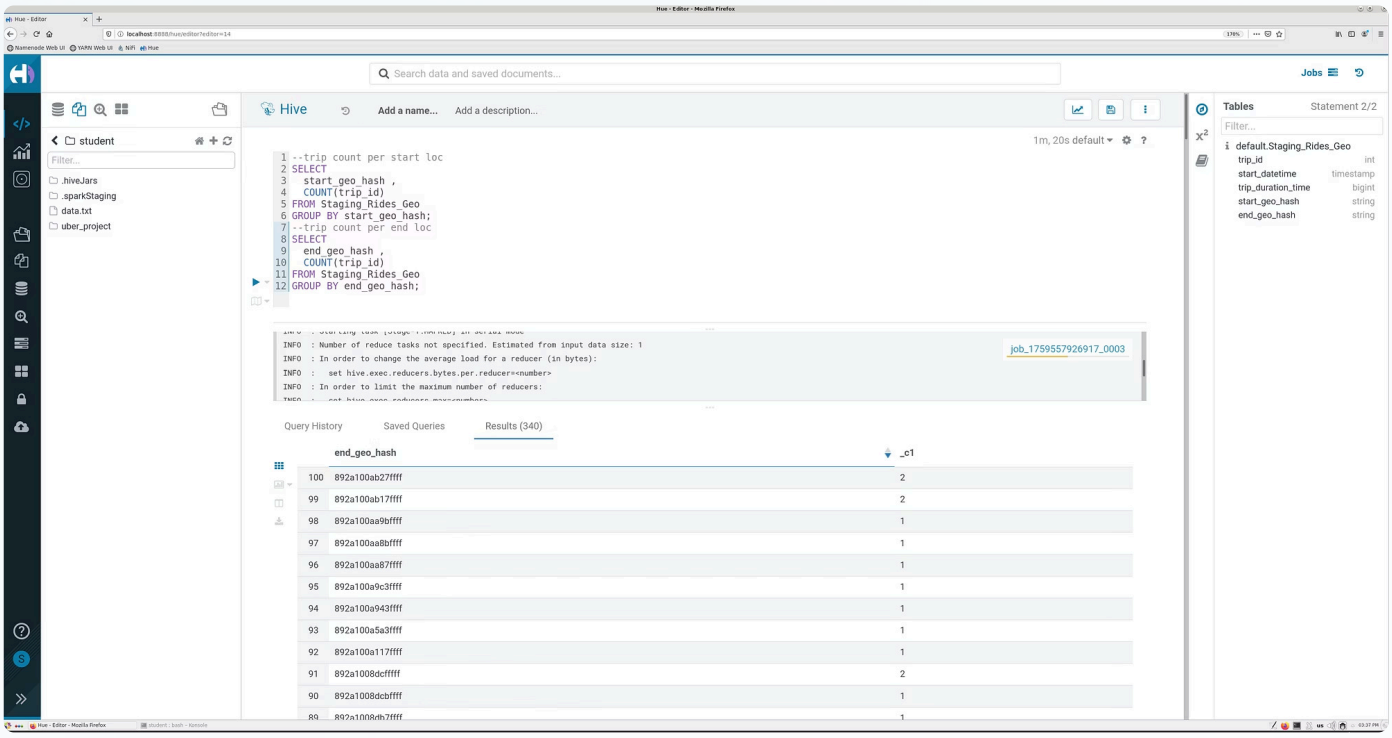


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1. Trip Count per Start Time

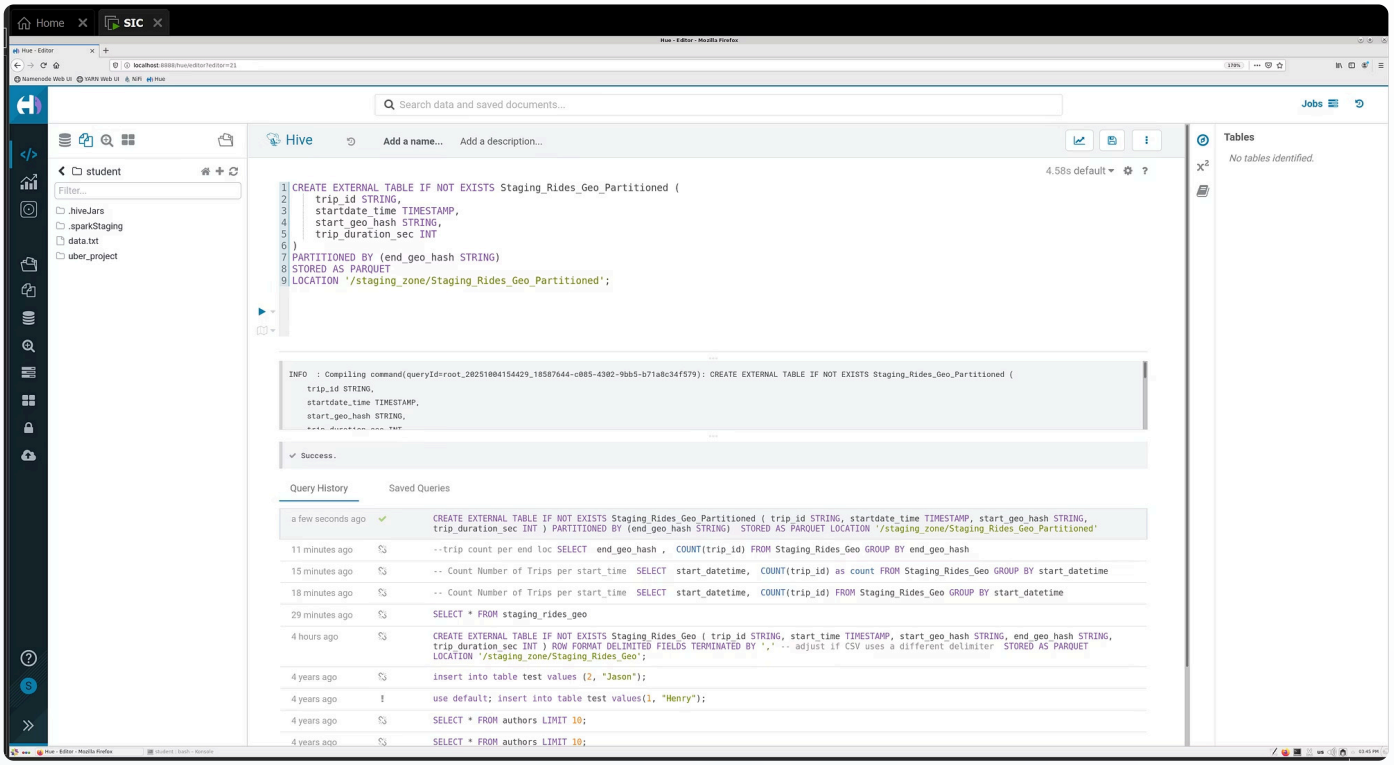


3.identify high-demand pickup points and identify popular destinations



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Step 4: Partition the Table



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